

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

**Course: B. Tech. Branch : Electronics and Computer Engineering / Electronics
and Computer Science Engineering**

**Semester :III Subject Code & Name: Computer Architecture & Operating
System (BTESC304)**

Max Marks: 60

Date:09-01-24

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Draw the connections between the processor and main memory and explain the basic operational concepts.	CO1	6
B) Explain the different kinds of addressing modes with an example.	CO3	6
C) Explain the Intel X86 architecture with a neat block diagram.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain briefly about Associate-mapped and set-associate mapped cache.	CO2	6
B) What are the disadvantages of single continuous memory allocation? Explain.	CO2	6
C) Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6 How many page faults would occur for the optimal page replacement algorithm, assuming three frames and all frames are initially empty.	CO4	6
Q. 3 Solve Any Two of the following.		12
A) Explain the basic concepts of instruction pipelining in computer architecture. How does instruction pipelining improve CPU performance?	CO3	6
B) Draw the microinstruction-sequencing organization of next-address field and explain it.	CO3	6
C) Explain input/output operations of computer architecture.	CO3	6

- Q.4 Solve Any Two of the following. 12**
- A) What is the difference between a process and a thread? CO4 6
- B) Define the Operating System. Describe the operating system's objectives and functions. CO4 6
- C) Assume the following workload in a system: Draw a Gantt chart illustrating the execution of these jobs using Round robin with time quantum = 2 unit scheduling algorithm and also Calculate the average waiting time and average turnaround time. CO4 6

Process	Arrival Time	Burst Time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

- Q. 5 Solve Any Two of the following. 12**
- A) What is the critical section? What are the minimum requirements that should be satisfied by a solution to critical section problem? CO5 6
- B) What is Producer Consumer problem? How it can illustrate the classical problem of synchronization? Explain. CO5 6
- C) Explain deadlock avoidance using banker's algorithm with suitable example. CO5 6

*** End ***